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Electro-Mechanical Integrators, Inc.
IDEAS INTEGRATED WITH TECHNOLOGY

Instructions and General Definitions For TPC's and PieceMaker

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This reference manual is meant to be read and searched in PDF form for keywords and subject matter. However, reading the entire manual which can be done in under an hour is required for every programmer and operator prior to the start of training. It is not necessary to memorize any part of it.

PieceMaker is an EMI-Registered Trademark and Copyright. The software's strength is that generally anyone involved in the industry who is adept at reading prints or who can sketch a skeleton of the structure with a pencil can quickly use the software to get into production with circular cross section parts (tube and pipe). The software not only creates geometry but handles other aspects of running the machine such as torch control, lift management, auto-loading, and part nesting. This is particularly helpful for companies without engineering resources who are just interested in making pieces from sketches or supplied prints. Many of our customers fit into this category.

There are some optional import capacities. Some of the importing abilities include importing cut lists of previously programmed parts. Importing Piece- and Batch-level data from Tekla via STP files and SigmaTube or Lantek. Importing all data including piece level data can be done at the Access database level via Excel or the other Office standard file formats by customers with the right resources. Please research this subject thoroughly prior to purchasing a TPC, PieceMaker, or other related software.

The industry is awash with different ideas and standards on interoperability between software packages and or machines. Here are some specifics that apply to handrail manufacturing and these machines. For example, the DSTV (NC1) standards do not cover round material assemblies. This is further complicated by the fact that handrail is often not detailed as if it were going to need valid geometry for export. Similarly, even the CNC files that most steel packages export do not adequately cover the geometry of the pipe joints and again there is no standard for breaking down assemblies or complete railings into useable parts. Further there is a lot of work done in handrail shops that only resembles the original prints. How joints and elbows are done and applied is a shop-by-shop preference and is never adequately represented in the models. Weep holes and field splices are rarely represented in the models. It is for these reasons that the team on the floor will need to modify many of the pieces that would be imported anyway. We at EMI caution prospective buyers that there are no magic wands. Please research these matters thoroughly prior to purchasing a TPC, PieceMaker or other related software.

Definitions and Reference Information Arranged Alphabetically
Please note not all features or definitions are installed, available, or applicable on all machines.

Abbreviations, common:

C90-C90 : A piece with a 90 deg cope on both ends. A typical mid-rail.
F45-F45 : A piece shaped like a gusset with a 45 deg flat cuts on both ends.
C60-R180-C60 : A piece with 60 deg copes on both ends with a 180 deg rotation between them. A typical mid-rail on a stairway or a diagonal in a truss.
C60-F90 : A piece with a 60 deg cope and a 90 deg flat cut.

Auto-Support: Most 2100-series machines are equipped with a pair of pneumatic devices that support the length of the material until the chuck and carriage approach their positions. At that point the material is typically able to support itself. Active in both manual and auto-loading routines, they automatically fold out of the way when the carriage approaches. They should not be confused with the Material Lifts that are part of the auto-loading process on the 2400-series machines. The height of the Auto-Supports gets adjusted with a short sample of the material to be cut in place and the air turned off such that one can physically raise and lower the lift.

1. Jog X positive and put supports up.
2. Turn off air at machine inlet.
3. Loosen adjustment lock lever on rear auto-support ("Lift 3) so it drops down.
4. Jog X negative until set-up stub is above the support, very close to the chuck.
5. Manually raise support V-block until it touches material, then drop it about 1/8" and lock it.
6. Push that auto-support down out of the way.
7. Jog X negative and repeat with front auto-support ("Lift 2").
8. Turn air back on and proceed.

A practical height for these is approximately 1/8" below the actual contact point such that there is some allowance for bends in the pipe to belly down without lifting the front of the material out of the front V. At a practical level Auto-Supports set for 1-1/2 pipe might not need to be reset for 1-1/4 pipe. Please note that the Auto-Supports are numbered #2 and #3. They are both actuated on AUX-3. They are individually actuated with AUX-6 and AUX-9 respectively. There is no #1 nor #4 like there is with the Lifts on the 2400-series machines.

Batch: A number of parts nested and organized for manufacturing. The Batch can contain multiple different pieces if they are all the same diameter and wall thickness. The Batch exists

in parametric form in the database like the Pieces. The Batch is processed into G-Code only when the G-Code button is pressed. See Nesting for more details.

Bend Notch: This is a software option released early in 2019. Most machines can take advantage of it with the appropriate software upgrades. A bend-notch is programmed like the other features. A quick hint would be that a straight pipe is 180 degrees. The programmed bend-notch angle is subtracted from 180 degrees and what is left will be what a protractor measures the actual angle to be. For example, if you program a 90 deg notch you will get a 90 deg angle in the part. If you program a 30deg notch you will get a 150deg part, typical in many sloped handrail top rails.

Chuck: A manual chuck generally used on the 2100-series and some older 2400-series machines. It has a standard chuck key, like a lathe or drill chuck, and is for round material only. The TPC does not produce high torque, so generally tightening with one hand in one key-hole is sufficient for OD-gripping smaller material profiles. When ID-gripping larger materials it is recommended to apply two hands of torque.

When seating a piece of material in this chuck, it is good practice to jiggle the material around between the chuck jaws while tightening. This reduces the chance of accidentally clamping the material with only two jaws instead of all three, which will mis-align the material and it is likely the material will fall out, during the program. Keep this chuck wiped clean. Keep the key in its holder built into the carriage, so it is not misplaced or broken when the chuck rotates.

Cope: The shape put on the end of tube or pipe so that it mates up to one of a same or larger diameter. Other industry terms include fish-mouth and notch. PieceMaker expects Center-to-Center lengths (dimensions) for all parts and will calculate and subtract whatever is necessary for the copes to fit properly with the total length. The actual calculated lengths of each part can be seen in the Batch window.

CR-A: Center of Rotation of the A-axis. Please note that this is a fixed point on the machine and serves as a reference for multiple other actions and setups. In most cases Z-axis is referenced to the CR-A as 0.000. Once set, this value should remain fixed as it cannot physically change. A precision rod is used at the factory for the setup process. There are values related to this that are used in other low-level parameters that influence velocity calculations. Please see related material on the Front Material Support section.

CR-Y: Center of Rotation of the Y-axis. Please note that this is a fixed point on the machine and serves as a reference for multiple other actions and setups. In most cases Y-axis is referenced

to the CR-A as 0.000. Once set, this value should remain fixed as it cannot physically change. A precision rod is used at the factory for the setup process. There are values related to this that are used in other low-level parameters that influence velocity calculations.

Database: All the values entered into PieceMaker are stored in a Microsoft Access database file located by default in the C:\Documents\PieceMaker directory. There can be multiple databases stored and used to organize work. Moving bulk data in and out of the database can be done in multiple ways including optional and custom solutions offered by your vendor. The recommended directory for the database is C:\Documents\PieceMaker. In situations where a company is organizing work for Job #1 while Job #2 is still running, we recommend that a second database be started. Expanding that thought, it could be stated that each job should have its own active database such that when production runs overlap there is better chance of having common parts retained and less chance of confusion where part numbers might be similar.

Flat Cut: An option to make cut geometries like that of a saw cut. Sometimes called a miter cut. Be careful when working on prints that use long point dimensions because PieceMaker is looking for dimension to the center of the tube like it does on the coped parts. Please note that the joining diameter changes to the default Flat Cut Join Diameter in the PieceMaker parameters page when a flat cut is chosen. The advantage of PieceMaker's Flat Cut over a saw cut is that the knife edge is eliminated, weld prep time is reduced, weld quality is increased, and finally grind time is reduced. The actual shape of this cut and thus it is weldability can be adjusted in the Parameter Page => Operating Parameters => Flat Cut Join Diameter. The typical range is 18-200. Handrail fabricators seem to like smaller numbers for the miters. 50 works well.

Front Material Support: See Material Support

G-Code: An international standard of programming for machine tool controls. There are some variations among manufacturers. Please see Centroid's Manual for a full listing. The G-Code button in PieceMaker converts the Batch data into machine ready G-Codes and M-Codes. Other ways of generating G-Code for the machine include Intercon, SigmaTube, Lantek, and macros. For most users and applications manual coding is impractical and little knowledge of G-Code is ever required.

Gripper: Pneumatic gripper on some 2400-series machines. It has replaceable jaws, which are swapped out to match each type of material profile. Some jaws cover a range of sizes, some

are dedicated to one size of material. They hold the material in such a way as to match the Front Material Support, and to match how the part program is expecting to cut the pieces. They swap out in a minute or two and are one of the five set-up considerations whenever changing material profiles on the 2400-series machines.

Homing: Homing is the process by which the machine finds its initial reference positions upon being turned on and told to do so. There are two basic ways it takes place. When powered up the machine says, "Warning Machine Not Homed". If you press cycle start at this point the machine will proceed according to the Home File. The home file acts like any other *.CNC G-Code file. You can see it on the screen during the process. It usually moves Z+ first and then follows with the other axes in a sequence that minimizes the opportunity for a crash. Always remove any material in the chuck before homing. A common failure mode for the process is dirty and or misadjusted home switches. The machine does not rely on the precision of the actual switch for the repeatability of the process. It uses the encoder index pulse for that. Some machines do not use conventional switches. Instead they watch for a quick spike in current when the axis hits a hard stop and then reverses to an index pulse for a precise location. If an error pops up on the screen while homing that talks about the number of counts changing you want to answer "N" and jog the axis away from the end-stop and try again. If it happens a second time, please notify your service organization. Answering "YES" may or may not cause an absolute position shift which often does not prevent operation but may cause unexpected interference issues. Again, please consult your service organization if you see this message. The homing process can be repeated at any time without power cycling the machine by loading the CNCM.HOM file out of the C:\cncm directory and running it like any other *.CNC file.

Lift-1: The automated lift closest to the torch or front of the machine. This lift is handled two different ways. First, on the 2400-series three-axis round-only machines where the material runs on at the torch on a V-notched glider Lift-1 should be set approximately 1/8"-3/16" (3-4mm) above CR-A so as to have the material clear the front support as it is feeding forward during loading. The lift is then automatically lowered and is not used again in standard operations. In the second case, the material is being supported at the front of the machine by a rotating assembly (the most common). In this case Lift-1 should be set so the material is right on CR-A or at most 1/16" or 0.06" low. If the support is significantly high or low the material will hit a bearing assembly. If it is a little low as stated above, you will observe a slight lift as it enters the carrier. If it is a little high the lift will force excessive deflection into the carrier springs.

Lift-2: The second automated lift from the torch or front of the machine. This lift's primary purpose is participating in the material loading routine. It should be set approximately 1/8" (3mm) below CR-A. Having this lift set low helps the machine deal with bent tubing. It can be set as low as 1/4" below CR-A but lower settings on some materials may introduce a sag transition from the time the stock material transitions from long to short. This transition can be seen in the Flame Distance. This lift is sometimes used to support non-round profiles with a cyclic movement matching the short and tall side of material without interfering with the rotation process.

Lift-3: The third automated lift from the torch or front of the machine. This lift's primary purpose is participating in the material loading routine. It should be set approximately 1/8" (3mm) below CR-A. Having this lift set low helps the machine deal with bent tubing. It can be set as low as 1/4" below CR-A but lower settings on some materials may introduce a sag transition from the time the stock material transitions from long to short. This transition can be seen in the Flame Distance. If short stock is being used in the 2400-series machine and Lift-4 is not in use then Lift-3 should be set right on CRA because it is now controlling the position of the end of the material as the Chuck comes forward to grip it during loading. M82 needs to be modified to run auto-loads on material shorter than 24ft.

Lift-4: The fourth and last automated lift from the torch or front of the machine. This lift's primary purpose is participating in the material loading routine. It should be set right on CR-A because it is controlling the position of the end of the material as the Chuck comes forward to grip it during loading. People who have a 24ft machine running short material in auto-load mode may want to comment the lift-4 management out of the M82.

Lift Cap: Each pneumatic lift has a replaceable cap. Different material profiles often require different lift caps. There are various cap designs for various material profiles. Some material profiles require different caps, and lifts set to different heights, for the material-handling functions to work. Some caps are specialized rollers, others are sacrificial polymers that wear and are replaceable. Lift caps generally take seconds to a few minutes a piece to change out. You may or may not have to change lift caps, when setting up for a different material profile. This will be covered in training.

Lift Height: On most 2400-series machines, the auto-lifts have adjustable stops to control the height of each individual lift. Some material profiles use all the same Lift Caps and Lift Heights, others may use different Caps and Height settings. Each lift has a lock knob, an adjustment nut, and an indicator display. To adjust the height of a lift, loosen the lock knob, use the factory-

supplied speed wrench to dial the adjustment nut up or down as needed, referencing the indicator's displayed number. Once height is set, lock the knob again. Some material profiles require different caps, and lifts set to different heights, for the material-handling functions to work. This will be covered in training.

Load Steps: Load steps are primarily used on 2400-series machines with auto-loading. On the 2100-series machines without auto-loading they would usually all be set to zero (0.0). Load Step-1 is how far the X-axis feeds until the jaws close. Load Step-2 is how far the machine feeds until the material enters the front support/carrier and Lift-1 is automatically lowered. Load Step-3 is how far X-axis feeds until the material is under the torch and ready to burn. Load Position minus the three Load Steps must equal the Stock Length setting in the Batch window. Operators generally should not have to adjust these Load Step settings.

M-Code: An international standard of programming for machine tool controls. There are some variations among manufacturers. Please see Centroid's Manual for a full listing. EMI's TPC has some custom M-Codes. They are listed in the appendix and can be seen in the C:\CNCM directory. You should not have to learn these or use them in typical operation. DO NOT MODIFY THEM. If you are programming or configuring your machine for special case operations, you may reference them. EMI's engineering staff can help with these situations.

Material Support: The support mechanism nearest the torch is the called the Front Material Support. It is not one of the Lifts talked about elsewhere. There are two types. One is made up of a V-block and usually works in conjunction with a similar device called the material Hold-Down. This V-block Material Support is adjustable in a vertical direction so that the center of the material matches the center of rotation of the A-axis or CR-A. Adjustability in the Y-axis is there such that the center of rotation for the material is similarly aligned in the Y-axis with CR-A. It would be unusual to adjust the Y with material size.
The second type of Front Material Support is a rotating assembly that is sometimes adjustable to fit various sizes of material and other times it is a dedicated plate ("puck") with fixed rollers that match up to specific material profiles. Most commonly this rotating assembly is driven with a servo motor and electronically slaved to the A-axis. The rollers do not wear like the polymer blocks and the material is captured in the Y-Z plane such that there are no location adjustments for material wear or material size. On machines equipped with Auto Lifts the setup for Lift-1 changes depending on whether or not the there is a V-Support or a Rotating Front Assembly (Tombstone and Puck).
The Adjustable Puck gives some flexibility in the standard tooling package on the 2400 machines. However, in situations where the shop runs a lot of a particular material profile, it

may well be preferable to get a dedicated puck for that specific material, rather than adjusting the adjustable puck each time. The support pucks swap out with 2-4 bolts, and self-align on pins, in minutes. The Adjustable Puck adjusts in minutes, with 4-6 jaws, each with one bolt.

Miter-Joint: See Flat Cut

Multi-Joint: An end type that fits into an existing joint of two other elements. When programming this in PieceMaker, there is a requirement that the three elements at the node must be co-planar and have a common intersection point. If this is not the case the part will not fit correctly. If you have complicated assemblies modeled in 3D beyond PieceMaker's capabilities, please check out SigmaTube and/or Lantek third-party software packages

Nesting: Nesting is the process of optimizing material utilization. Indeed, when applying the concept to conventional flat worktables the parts might even be interlocking. In PieceMaker, if you put the actual length of the material loaded into the stock length window and the Stub Length is set appropriately in the Parameters window, you can start loading desired pieces into the batch and watch the remaining length count down. Without using a sophisticated algorithm or third-party software it is good practice to simply work longest to shortest. Working longest to shortest will typically deliver the added benefit of preventing a very short length of material being in the jaws with a long length hanging out past the front support and over-torqueing the gripper/chuck assembly. See ROP for more critical information.

Networking: All current machines are being shipped with Windows 10. If it is your intent to put the machine on your corporate domain, it may to need to be upgraded to Windows 10 Pro. All machines released after approximately 2012 using the Windows-7 OS or later can be networked using TCIP protocols. Many earlier machines can be networked also but the procedure gets more involved. EMI does not provide IT services. Networking is always the responsibility of the customer. EMI staff will provide basic instructions for the machine side but will not under any circumstances work on the customer's network equipment or PCs. Be warned: while the IT staff is doing network setup it is possible to disable the machine in multiple ways. Common ways of disabling an otherwise working machine include reassigning IP addresses incorrectly, enabling fire walls, and installing anti-virus software. EMI is going to charge for the work of restoring the machine to functional at a Time-and-Material level without warranty, double if your IT guy is egotistical, maniacal, or impolite. The customer's IT staff would do well to talk to our staff prior to attempting anything.

Part Setup: Chapter 4 in the Centroid Manual covers this subject in general and in depth. However, the plasma machines we are discussing here can be setup with the following simplified procedure.

On the 2100-series, there are three steps to consider, any time an operator wants to change material OD:

1.0. Auto-Support Height setup (see Auto Support section)
2.0. Front Material Support Height (see Material Support section)
3.0. Z-Height/Torch-Height (see below)

2100-Series Machine X-Axis Set-Up (done at the factory, this should only need to be redone if someone mistakenly resets it. It is NOT a normal part of changing material diameters):

1.0 Remove any material from the chuck. Jog the X-Axis all the way to the torch end of the machine.
2.0 Be sure WCS#1 (G54) is displayed in the upper left-hand corner of the screen. If not use F6 or F7 in the Part Setup menu to switch it to WCS#1 (G54)
3.0 Jog the torch (Z-Axis) down near CR-A.
4.0 With a tape measure, measure from the face of the chuck (not a jaw, not in the center bore) to the center of the nozzle on the torch. Take the measurement twice and be sure it is accurate to better than 1/16". Convert it to its decimal equivalent and write it down. (On machines shipped since late October 2018 this value should be 10.00in.) This value is critical to a feature called ROP (Reverse Order Processing). Please read that section as well.
5.0 Now from the base level menu press the following sequence
    5.1 F1-Setup
    5.2 F1-Part
    5.3 Note that by default the X-axis is selected.
6.0 Enter the value from Step 4 in the first field
7.0 Enter 1.0 in the second field
8.0 Using the space bar to toggle the value in the 3rd field, Approach From, to "Center" if it is not already.
9.0 Press F10-Set
10.0 Note that the X-axis DRO now displays the number you measured. If not try again.
11.0 Now hit ESC a couple of times back to the base level menu.
12.0 Open PieceMaker and go to the Parameters Page.

13.0 On the center tab, Machine Parameters, the 3rd value down is Stub Length.

14.0 Enter the value from step 4 there and hit SAVE.

2100-Series Machine Z-Axis Set-Up:

1.0 Place a 16" piece of the round material to be cut in the chuck and tighten it up. Rotate A-axis rapidly and confirm that the set-up stub is true in the chuck, not wobbling. Carefully set the front V-Support such that it lightly contacts the material but does not force it up. Lower the material clamp such that the material stays in is snug down onto the front V-Support.

2.0 Jog the torch down to the Hypertherm recommended clearance for the consumables and material being cut. For most handrail applications this is 0.080"- 0.100". Use a known gauge block. Also read the section on **Torch Distance**.

3.0 F1-Setup F1-Part F1-Next Axis (repeat as necessary) until Z-Axis is displayed.

4.0 Now enter ½ of the material's known diameter. For example, 1-1/4 pipe has an O.D. of 1.660. Half of this is 0.830. If you are cutting 1-1/2 pipe it has an O.D. of 1.900, enter 0.950 into the field. Press ENTER. Now enter "1" in the tool field, (99.9% of the time these machines use only Tool #1). Hit ENTER again. Now press F10-Set. Observe that the Z-axis DRO now reads the appropriate 0.830 or 0.950. Also, at the tope right of the screen, near the Job Name, should display T1 H1 (Tool 1 Height Offset 1). If not try again. In later versions of software you can use the data field as a calculator. If you enter 1.66/2 "Enter" the field will calculate the answer (0.830) as expected.

5.0 The Y-axis center is setup mechanically. There is no CNC setup for the Y-axis on the 2100- series machines.

2400-Series Machines Part Set-Up:

On the 2400-series, there are up to five steps to perform, any time an operator wants to change material profile/size:

1.0. Lift Caps

2.0. Lift Height setups

3.0. Front Material Support Puck (Adjust the Adjustable Puck or swap Puck)

4.0. Z-Height/Torch-Height

5.0. Gripper Jaws

See each feature/function for detailed instructions.

This process is just like the 2100- series machines with the simplification that the height of the material is always controlled by the front carrier bearings in the puck. The Y-axis needs to be zeroed on the center of the A-U axis assembly. At a technical level, these values would never need adjusting because the center of the A-U axis assembly never changes. At a practical level we check and adjust them because the material is never perfect and the machine, specifically the torch, regularly gets inadvertently hit and knocked out of location. Setting up the Y-axis zero uses the same keystrokes as the Z-axis process but select the Y-axis instead of the Z in the final step. There are multiple ways to find the center of the material including just doing an eyeball job on round material. Using the torch body as an edge finder and following the procedures in chapter 4 of the Centroid manual is another. When cutting square, rectangular, channel and, angle-iron, and flat bar, the Y-axis centering process becomes more critical.

Piece: A singular part number to be manufactured. If programmed in PieceMaker it only exists in a parametric form in the database. Another way of saying this is that the piece only has the critical characteristics as entered in the PieceMaker fields saved. There is no G-Code associated with a Piece prior to pressing the Quick Code button or processing the Batch. With the highest priority, Pieces should be created with Flat Cut 90s on End 2. In second priority, 90-degree copes should be placed on End 2. Later, this practice will help with material utilization.

Printing Head: Some of our machines are equipped with a printing head. In most cases this is a self-contained Dot-Peening device. There is a group of custom G- and M-codes that controls this device. PieceMaker has an option checkbox in the Batch window that becomes visible when a default directory is set in the Database tab of the Parameters menu. Blanking the default directory field will remove the option checkbox from the Batch window. SigmaTube has a control set up for it that is visible during the post process event.

Slaved Axis "U": Most of the 2400-series machines are equipped with a driven front support puck. This axis is called "U". During most operations A-axis and U-axis are slaved together. Meaning Axis-U will follow exactly the moves of axis-A without specific calls on each line of G-Code. This slaving action is usually enabled inside the homing file after the home sequence is complete. Slaving remains enabled through power cycles but there is no way of tracking axis motion during a power off state, therefore slaving is first turned off in the Home file, then the homing sequence takes place and then slaving is re-enabled. Slaving enable is done with "G10 P64 R64". Enabling and disabling slaving can be done from MDI with "G10 P64 R0". With slaving disabled the spindle speed + and – keys jog the U-axis independent of A-axis.

Torch Distance: This is the distance from the copper nozzle of the torch to the material surface. It is considered part of the torch during Z-axis set up and programming. The recommended values can be found in the Hypertherm manual specific to your unit. This manual is shipped with the machine, and there is often a link on the machine's desktop, it likely can be found in the typical windows documents directory and always available on the Hypertherm Website. Experience tells us that 0.080"-0.100" (loose fitting on a nickel) is a pretty good setting. Note that the flame tip is the actual bottom of the tool--not the copper nozzle.

Tool Setup: Sometimes also called TLO. This subject is discussed at length in Chapter 5 of the Centroid Manual. However, for all practical purposes we do not use this feature and thus all the TLO values are zero (0.000). The exception is for 2100-series machines with scribes. In this case TLO 200 is used to differentiate the height between the torch and the height of the extended scribe.

Weep Holes: A common name for vent holes placed in welded structures that will be hot-dip galvanized. These vent holes keep sealed chambers from exploding and let the zinc solution coat the inside of the structure. Every shop has their own ideas when placing these vent holes. The basic tool will put a hole wherever desired. We also developed an automatic recognition system for handrail mid-rails both straight and on pitch. When either of these two types of pieces are recognized in the Create Piece window a check box appears below the Rotational Offset field. When checked the software will automatically place weep holes in the Piece definition according to the customer's preferences defined under the Weep Hole Defaults on the Features page.

Work Coordinate Systems: (WCS#xx) This section is a recommended standard such that EMI's tech support group might more quickly recognize certain issues and help customers get back into production more quickly. The standard machine comes with G54-G59 such that if you want to experiment or organize work in some unusual way you can start with G56 and work up from there. The Centroid Manual indicates that G54 is equivalent to E1 and G55=E2 and so forth for the first 6 WCSs. After that it continues as per the specifics in the manual. Generally, we use the "G" version though at the time of this writing we are considering the potential value in the differential visibility of using the "E" notations.

G54: On a TPC, G54 is used for round work coming out of PieceMaker. There are two distinct ways to set the X-axis part zero. Sometimes, G54 is zeroed with the chuck at the negative limit of travel or nearest the torch and not changed during operation. That means as the machine processes the material down to the Stub Length the DRO will be reading 0.000,

even though there may be 10 inches of actual material left in the chuck. The 2nd way to set the X-axis part zero is to jog the carriage all the way to the torch end and measure from the face of the chuck to the centerline of the torch nozzle. Setting this value in the X-axis registry now gives you a frame of reference from the torch. This value is required for ROP operations and must be entered into PieceMaker as the Stub Length.

Y-zero is the center of the material as it is ultimately referenced off the CR-A, a fixed point. Z-zero is the center of the material as it is ultimately referenced off the CR-A, a fixed point. A-zero is often set such that the jaws are vertical so that in case of an air pressure loss or unintentional jaw opening the material does not fall from the open gripper, but just sits on the lower jaw. U-zero usually has some useful correlation to A-zero although their respective offsets from machine zero may be different due to tooling and mechanical assembly differences.

G55: On a TPC the G55 WCS and above is often used for square, rectangular, angle iron, C-channel, and flat stock. The X-value is often being reset by G92 calls from each piece or processed group as they have their own localized zero. SigmaTube and possibly the other CAM packages put out only Absolute coordinates. Y-zero is the center of the material as it is ultimately referenced off the CR-A, a fixed point. Z-zero is the center of the material as it is ultimately referenced off the CR-A, a fixed point. A-zero is set such that the jaws hold one face of the material horizontal, parallel with the Y-axis. U-zero usually matches A-zero though their respective offsets from machine zero will be different.

G56: On a TPC G56 is sometimes used for all material but in a manual kind of mode. The X-value is usually fixed and referenced with the Stub Length included. Different than G54 when the machine is sitting at the negative end of travel the X-value will display the stub length. Indeed, anywhere along the machine's travel the X-value displayed is the actual length of material in the machine. When used in conjunction with custom Cut-Off Macros or Subroutines the machine can be used as a somewhat automated saw. This mode of operation is sometimes used to cut good pieces that include the Stub Length Material that might otherwise be scrapped.

Part Height Setup: Also called Z-zero setup, or torch set-up. This topic is related to the Work Coordinate System explanation. In every case, except perhaps when cutting flat stock, Z-zero is CR-A. It is important to grasp that while the surface height is changing for different material profiles, once set, the Z-zero value should not change for any material being rotated because obviously CR-A does not change. The control as programmed from PieceMaker and SigmaTube accommodates the material diameter both real and effective as in the case of square and

rectangular shapes. Note that the machine position is set with the radius not the diameter. Similarly, when the machine is programmed conversationally with PieceMaker diameters are used in the fields, but note, the G-Code comes out in radii. Please note that when the Material Front Support is a V-block that the center height of the material requires a non-linear adjustment from CR-A. In practice, the center of the material on 2100series machines and the torch height is something that needs to be watched and often is simply adjusted on every material diameter/size change over. It only takes a minute and accounts for other real-world conditions.

Quick Code: The Quick Code button on the Modify/Create Piece page is a short cut for creating G-Code for a single piece. The button creates the Batch and G-Code with the Piece name as a Batch name and a "-QC" appended to it. The Quick Code button does single pieces only and does not include an Auto-Load Sequence. It uses the default feed rate parameter value (named Between Cut Rate in Parameter window).

ROP: Reverse Order Processing. This feature reverses the batch layout such that the end of the last piece still in the chuck and the length of material that historically was wasted as the stub length now can be used as a finished end on a piece that has a 90F programed as End 2 (provided the customer is willing to use the stock material's factory end). Additionally, this pushes all the unused material on a length to the front of the stick. There are three (3) triggers that enable this feature. The first is a variable in the machine parameters section of PieceMaker and defined lower in this section. The second is setting the ROP check box in the batch window. Thirdly, setting the Auto-Load check box to true. The variable value can be an integer between (as seen in the manual under user variables) between 29000 and 31999. This value will appear in the G-Code as part of the load routine. A value of zero (0) turns off the function. Don't change this unless you are really getting serious about creating some special functions. Current machines leave the factory set at 29000. Depending on your batching needs and preferences you may turn it on and off.
In order to use the ROP feature you need to set the X-Value in the G54 WCS in the following way. First jog the carriage all the way to the front of the machine then measure the distance from the appropriate hard stop in the chuck jaws to the torch center. That value is then entered into PieceMaker as the "Stub Length". When done correctly it would follow that the length of material in the machine after a straight cut off would match the DRO. Said another way, when correctly done the end of the pipe that you cannot actually get to would be 0.0 on the DRO if you could get to it. On machines with torches that adjust in the X-axis it would be correct to set the torch position to the smallest appropriate value for the material being processed and a convenient value to measure since the accuracy of this measurement is

precisely the accuracy of the last piece cut when using the ROP feature. Pay attention to where the material seats in the jaws. Depending on purchased options the actual seat position may change with material diameter. The measurement, the G54 offset, and the PieceMaker parameter "Stub Length" need to match for ROP to work. Currently ROP only works at an all-or-nothing level in the batches. If you want it on for any Stick it will be on for all the Sticks in that Batch. For this reason, some operators just go through the process with only single stick Batches. Sometimes this simplified approach pays other dividends as well. When creating batches, one will notice that pieces with a 90F on the 2nd end appear in dark green and in later versions have a gear beside them. For ROP to work the last piece on every stick in the batch obviously needs to be dark green, and have no features within the Stub Length

SIMUH: Something is messed up here.

Z-Zero Setup: see Part Height Setup

Custom M-Codes and Descriptions for EMI's TPC.

Not all machines make use of all M-Codes even though they may be in the directory. The customized codes may vary from machine type to machine type. Users may adapt codes to better suit their applications but do so at their own risk.
The purpose of this section is to be a reference of what they do, not to instruct a user on how to understand or modify the macro files. Custom M-codes are handled by macro files on the machine. For example **M3** = mfunct3.mac and **M15** = mfunct15.mac. For reference' sake M3 and M03 are equivalent.
In many cases the M-Codes work in pairs ie M64/M65. In most cases the even value will trigger the event and the odd will cancel it. There are exceptions where earlier standards needed to be maintained.
Some of the features have been mapped onto the Auxiliary Function keys of the pendent.

M3 Exists to make the torch operation compatible with some typical Cam packages. M3 calls M15 in the background and turns the torch on. Do not use it unless necessary.

M5 Exists to make the torch operation compatible with some typical Cam packages. M5 calls M16 in the background and turns the torch off. Do not use it unless necessary.

M7, M8, M9 Exist as part of the standard Centroid install and they manage the coolant system. M7 turns on Air Blow and M9 turns it off. M8 is not used on any machines as of this writing.

M10 & M11 Clamp and Unclamp the material hold down usually found up near the torch. Also, will fire the Scriber down/up on the 2100- series machines if so equipped. Often toggles on Aux 4.

M14 A special version of the M15, torch ignition, that does not wait for the cutting arc to transfer prior to permitting the machine to resume motion. It is primarily used on open profiles (angle and C-channel) where an edge start is required but due to material tolerance there may not actually be material near enough the pilot arc to transfer the cutting arc. In this case the machine just "goes for it" and the net gain in reliability is significant. There is a "/" in front of M14's and M15's. It makes the ignition subject to the control's Block Skip function found under the F4-Run Menu and enables the operator to do a dry run without the torch firing. Read the Centroid manual for further explanation.

M15 & M16 Start and stop the torch respectively. There is a "/" in front of the M14's and M15's. It makes the ignition subject to the control's Block Skip function. F4-Run Menu: F4-Skips ON and enables the operator to do a dry run without the torch firing. Pressing F4 a 2nd time will toggle the function OFF.

M17 & M18 These are used in the Scribing Operation to both extend and retract the scribe, lower and raise the scribe, and to apply a temporary G92 offset. You may note the M10 and M11 are used inside of M17 and M18. *** SIMUH***

M25 Raise the torch (servo- based Z-Axis) to the Z value of the G28 position. It is the equivalent of the Tool Check button on the pendent. This is a great standard Centroid feature and operators should take advantage of it to protect the torch during setups, manual loading, and interrupting a regular operation.

M31 Triggers the cam- based tube loader/indexer on the auto-loading machines. This is the equivalent of the AUX-4 button.

M32 Incorporates the M31 into a larger material load and stage routine. It requires customization for different length machines and should be considered a specialized tool for special events. Do not fool with it if you do not know what you are doing.

M44 Checks to see if there is material staged in an inbound chute so that the auto-load cycle may continue or to trigger a warning light if the machine is waiting for material in some automatic cycles. It incorporates M76/M77, an air blast, to blow off dirt such that so the sensor does not foul.

M46/M47 Lowers and Raises (engages and disengages) the optional printing head. The solenoid triggered is usually in the main bank with its own pressure regulator. This regulator controls the downward pressure of the print head stop on the material in the machine. There is a separate regulator that controls the amount of air pressure going to the needle in the print head. There are settings in the printer control module that control the density of the printing.

M54 Is a dummy file with no standardized purpose, but sometimes is involved in manipulations WCS.

M55 It is a complex code used exclusively on the Angle-Chanel Processing Center to calculate G55, the scribe torch offset, from G54 that was setup by the operator from the main torch.

M56 Is a dummy file with no standardized purpose, but sometimes is involved in manipulations WCS.

M57 Is a dummy file with no standardized purpose, but sometimes is involved in manipulations WCS.

M62 & M63 Raises and lowers lift #1 on auto -loading machines. This is the one nearest the torch. In standard operation the lift is lowered by the PLC from the background.

M64 & M65 Raises and lowers lift #2 on auto-loading machines. It is also used on short machines with only 1 lift to support the middle of small diameter material. In standard operation the lift is lowered by the PLC from the background. In some applications with non-round material, “wait for input-X” gets commented out. The 2100- series machines only have Lifts #2 and #3.

M66 & M67 Raises and lowers lift #3 on auto-loading machines. In standard operation the lift is lowered by the PLC from the background. The 2100 series2100-series machines have only lifts #2 and #3.

M68 & M69 Raises and lowers lift #4 on auto-loading machines. In standard operation the lift is lowered by the PLC from the background.

M70 & M71 Raises and lowers the torch if it is on a pneumatic Z-actuator. It sometimes incorporates the M10/M11. On machines with servo driven Z-Axis it has no practical effect on operation.

M72 & M73 Raises and lowers lift #5 or #6 on auto-loading machines equipped with them. In standard operation the lift is lowered by the PLC from the background.

M74 & M75 Open and closes the primary chuck.

M76 & M77 Turns the part sensor air blast on and off. Incorporated into M44.

M78 & M79 All part loading lifts up and down.

M82 A complex code that fires the cam indexer (M31), raises all the lifts raises the material clamp (M10) and raises the Torch (M70 and or M25).

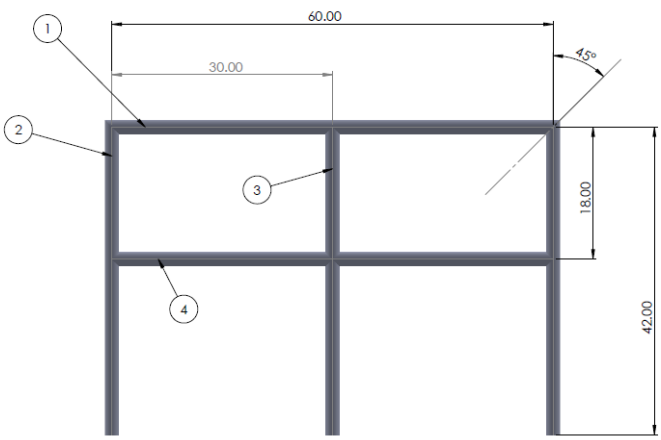
M84 A complex code/mode that enables a front-loading sequence on the 2100 and 2400 series machines when using PieceMaker 5.2.6 or newer.

M86 & M87 Extends and retracts a part ejector pin on some model**s.**

Aux Key Reference:
Aux 1: Chuck OPEN/CLOSE
Aux 2: Printer UP/DOWN
Aux 3: All Lifts UP/DOWN
Aux 4: Material Hold Down UP/DOWN
Aux 5:
Aux 6: Lift-1 UP/DOWN
Aux 7: Loader Indexer
Aux 8:
Aux 9: Lift-2 UP/DOWN
Aux 12: Lift-3 UP/DOWN
Aux 15: Lift-4 UP/DOWN

Programming in PieceMaker

Shortly following is a typical example of a popular handrail application. In the attached table are the dimensions required to program the parts and the names of the data fields where they go. All the dimensions for the pieces are used in the Piece Window to define the piece while the quantity of each individual piece is used in the Batch Window during nesting and so forth.



NOTE 1: THE PUROPOSE OF THE TABLE IS TO SHOW THE SIMPLISTIC INFORMATION REQUIRED TO PROGRAM PieceMaker SOFTWARE.

NOTE 2: THE MATERIAL IN THE EXAMPLE IS 1-1/2 SCH40 PIPE.

NOTE 3: THE END-1 IS CHOOSEN FROM THE UPPER LEFT CORNER OF THE PRINT. THIS ISN'T REQUIRED IT IS JUST GOOD PRACTICE.

NOTE 4: THE LENGTHS ARE INDEPENDENT OF MATERIAL DIAMETER AND WALL THICKNESS.

NOTE 5: ITEM #2 AND ITEM #3 HAVE THE LENGTH BECAUSE THE CENTER LINE INTERSECTION DISTANCE IS THE SAME.

NOTE 6: WHEN THE FLAT CUT BOX IS CHECKED THE JOINING DIAMETER IS CHANGED TO 200.0 AND DEACTIVATED.

TYPICAL DATA REQUIRED TO PROGRAM PieceMaker SOFTWARE

ITEM NUMBER	QTY	OD DIA	WALL THICKNESS	LENGTH	END-1 JOINING DIA	END-1 ANGLE	END-1 FLAT	END-2 JOINING DIA	END-2 ANGLE	END-2 FLAT	ROTATIONAL OFFSET
1	1	1.910	0.140	60.0		45.0	Y		45.0	Y	0.0
2	2			42.0		45.0	Y		90.0	Y	0.0
3	1			42.0	1.910	90.0	N		90.0	Y	0.0
4	2			18.0	1.910	90.0	N	1.910	90.0	N	0.0

Machine Operation Do's and Don'ts:

1.0 Do not smash the pipe or tube linearly into the chuck. The gearbox bearings can be damaged.

2.0 Do not apply bending forces to the chuck. This is not a reference to an A-Axis torque. Instead a bending force would be applied by leaving a long length of material hanging out of the chuck unsupported. Similarly, an opposite bending force can be applied when cutting a longer part and not supporting it with an outbound, outboard carrier such as a simple pipe stand. Obviously, the weight-per-foot and the length of the material both determine the force applied to the bearings in the rotating assembly. The limit for the 2100-series machine is 30 ft-lbs. This is the equivalent of 30lbs hanging on a 12" length of pipe sticking out of the jaws. The gearbox bearings are the limiting factor. On the 2400-series machines the torque limit for the front rotating assembly is 90 ft-lbs depending on jaw configuration. This is the equivalent of 90lbs hanging from a 12" long piece of material. The gripper assembly is the limiting factor on the 2400-series machines.

3.0 Do not turn off the main power without shutting down the software side of the control first. There is a Windows based computer in the cabinet. Turning off the power prior to doing a standard shutdown will cause file corruption as with every computer.

4.0 There is a surge suppressor and electrical noise filter in most every machine. If your machine does not have one it can be added in every case. They all have indicator lights and fault monitoring circuits. They should be inspected every six (6) months, after any known power surge, and after any known power outage. These devices are there to protect the more expensive and critical components in the control. It is expected that in time they will fail while absorbing the spikes and noise that would otherwise cause damage to the sensitive control components. If there is a question about interpreting these indicator lights or circuits, please contact your service provider. In many cases the machine will remain operational even after this component goes inoperable, however replacing it immediately is part of the warranty requirement and a wise investment.

5.0 If your shop experiences frequent power outages or brownouts, please talk to your service provider about having an industrial UPS (uninterruptable power supply) installed. Remember in almost every case people have their home and office computers plugged into a UPS and Surge Suppressor. How much more important is it to protect an expensive industrial tool in a nasty environment?

6.0 Do not modify the networking addresses and protocols assigned in the machine when it was delivered. The DHCP setting on the public port will permit most installations to access the internet for remote support. If it is necessary to put the machine on a secure network for file sharing and so forth, please consult your service provider first.

7.0 Do not adjust the tension on the main drive belt. The 2100-series machines and some 2400-series machines use a large, toothed belt to position X-Axis. These belts are tensioned at the factory and the machine is then calibrated. It is not a process that should need repeating until years down the road. Depending on model the belt can be at 1800lbs of actual tension or more. It requires specialized tools to handle it safely. If there is a reason to get involved with the belt please contact your service organization.

8.0 Do not install anti-virus software on the machine. It will slow down the Real Time Data Packets on its internal network to the point that the machine will show errors. If there is reason to scan the machine do it from a removable drive or a remote/network appliance. <u>Do not adjust the windows fire wall settings, they should all be off.</u>

Machine Maintenance:

There is very little required in the way of routine maintenance. Like every mechanical device the cleaner the moving parts are kept the longer they will last. The components on this system were selected and manufactured for years of trouble-free service within their expected service duties. Please keep in mind that oil and grease collect dirt, and in most cases, we have chosen materials that do better dry and dirty rather than caked with oxide-laden grease.

1.0 There are grease nipples on the bearings that carry the indexer shaft. A single squirt every 12-months will be good.

2.0 The Z-Axis linear actuator has a couple of sliding surfaces.

- 2.1 Brush the running surfaces with a nylon brush to clean off dust and debris every week or every 40 hours of operation.
- 2.2 The screw runs in a polymer nut that runs dry intentionally to avoid collecting the abrasive dust and smoke from the environment and torch. At the time of this writing the systems have been in the field for about five years and we have not seen wear nor had a failure. Brush with a nylon brush and blow the screw clean with compressed air every week or 40 hours of operation.

3.0 The Rolling-Sliding Material Support on the 2100-series machines.

- 3.1 Most machines do not have them.
- 3.2 The rollers/bearings are sealed and sized to last a long time.
- 3.3 The Sliding Material Support lifts right off the beam. On the underside it has plastic glides. Use a wire brush to remove any embedded debris. When the slides show significant wear replace them.

4.0 Auto Supports on the 2100-series machines. Not all machines have them.

- 4.1 The steel pivot axel has small grease ports that can be charged with the included grease syringe commonly recognized as the tool used to grease the carrier gear on the front of a chainsaw bar.

5.0 The Rack and Pinion.

5.1 If your machine has a rack and pinion drive it should be cleaned every year or as necessary. There is an auto-lubrication system inside the X-axis cover near the X-axis servo. It is estimated to have a 2-year supply of lubricant in it.

5.2 The lubricant is sticky and cleaning the rack takes some effort. We recommend a nylon brush and some mild solvent.

5.3 The lubricant applicator is a felt gear. It should be applying lubricant to the rack not dirt. If your felt applicator is black and caked with dirt it needs to be replaced. There is a possibility that the lubricator pump has stopped working. There is a 2-year life expectation on the battery and the lubrication reservoir. If you suspect that it is time to service or replace this unit, please contact your service provider.

6.0 The Pneumatic Material Lifts on the 2400-series machines.

6.1 The guide rods are supported by self-lubricating bushings. You can clean them but do not lubricate them. Oil or grease will mix with the dirt and jamb them.

6.2 Many oils and solvents will cause them to swell and then they will jamb solid.

6.3 They will last a long time if they are mechanically cleaned or cleaned with water and a mild soap.

6.4 The bushings are precisely bored to match the very rigid fork that fits in them. Replacing the bushings is not trivial and we recommend that they not be meddled with until something is significantly worn.

6.5 Brush the dirt buildup off of the top of the support rod bushings and the air-cylinder rod bushing every week or two.

7.0 Material Indexing Cams

7.1 The roller that lifts the arms rides in the groove in the side of these plates. Blow these out regularly. A small piece of debris will stop these in the middle of operation with surprising efficiency.

7.2 Do not lubricate these grooves.

8.0 All the pneumatic seals in the machine are made to operate without lubrication. Please do not add oil anywhere. There is not a factory oiler.

9.0 Quickly wipe the dust off the top and bottom of the 2100-series main beam before every use. The top and bottom corners are rolling surfaces.

AC-PC Specific Addendum

How to setup for cutting angle iron.

1.0 Machine Setup A-axis
- 1.1 Rotate A-axis until the Y-axis actuator is level using a quality level.
- 1.2 A-axis DRO should read 0.000 plus or minus 0.010 in the machine coordinates and the local G54 coordinates.
- 1.3 If not, the home file needs to be adjusted. Call your service organization.

2.0 Machine Setup W-axis
- 2.1 Bring the material to be cut into the machine.
- 2.2 Bridge the material from the loading conveyor to the unloading conveyor
- 2.3 Lower the Hold-Down
- 2.4 Lower W-axis with the U-axis button.
- 2.5 Install the orange centering tool on the A-axis.
- 2.6 Bring W-axis up until the tool touches the bottom of the material.
- 2.7 Use the supplied table to get the position value specific to the material being cut.
- 2.8 Enter that value into the W-axis position field. It should be negative and somewhere between 0.000 and 0.500
- 2.9 Remove the orange tool
- 2.10 F3 MDI: G00 W0.00

3.0 Machine Setup Z-axis
- 3.1 Press Tool Check
- 3.2 F3 MDI: A-45.0
- 3.3 The head will move counterclockwise.
- 3.4 Jog Z-axis down to about 0.100” above the material surface.
- 3.5 Set the Z-axis position to half the leg length.

4.0 Machine Setup Y-axis
- 4.1 With A and Y still in positions from the previous step:
- 4.2 Jog Y-axis positive to the heel of the angle iron and use a straight edge to align the center of the nozzle with the face of Leg-2

4.3 Set the Y-axis to the positive value of half the length of Leg-1

4.4 If you jog Y-axis negative at this point the center of Leg-1 should be Y0.000